

Learning Intention: I can estimate the number of objects in a collection and check if a calculated answer is reasonable.

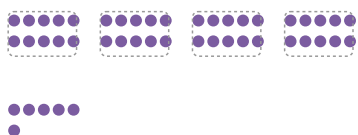
Success Criteria — I can:

- Estimate how many objects are in a collection by grouping.
- Round numbers to make an estimate before calculating.
- Use my estimate to check if an exact answer is reasonable.

Strategy Box — Two Kinds of Estimating

Estimating a collection	Estimating to check an answer
Count one small group, then use it to judge the size of the whole collection.	Round the numbers in a problem to friendly numbers, then calculate roughly to predict the size of the answer.

Worked Example 1 — Estimating a Collection



1. Look for a group you can count quickly — here, one group of buttons is **10**.
2. Count how many groups of about that size fit in the whole collection — there are about **4 full groups**, plus a smaller group.
3. Estimate: about $4 \times 10 = 40$, plus a bit more — a reasonable estimate is **about 45-50**.

Worked Example 2 — Checking Reasonableness

Problem: $38 + 25 = ?$

Step 1: Round each number to the nearest 10: $38 \rightarrow 40$, $25 \rightarrow 30$.

Step 2: Estimate: $40 + 30 = 70$.

Step 3: Calculate exactly: $38 + 25 = 63$.

Step 4: Compare: 63 is close to the estimate of 70, so the answer is **reasonable**.

Review: Rounding to the Nearest 10

If the ones digit is 5 or more, round up. If it's less than 5, round down. E.g. $38 \rightarrow 40$ (8 rounds up), $25 \rightarrow 30$ (5 rounds up), $22 \rightarrow 20$ (2 rounds down).

Try It

Round 47 to the nearest 10: _____ Round 63 to the nearest 10: _____

Parent Tip

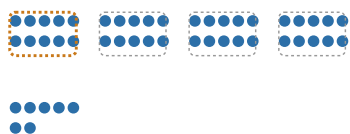
Ask your child to estimate before counting exactly — "About how many grapes are on this bunch?" or "About how many steps to the letterbox?" Then count together to check how close the estimate was.

Key Vocabulary

Word	Meaning
Estimate	A sensible guess based on reasoning, not an exact count
Round	To change a number to the nearest 10 (or other friendly number)
Reasonable	An answer that makes sense compared to your estimate

Build Skills — Estimate collections and check reasonableness.

Q1 — Estimate the Collection



a) One group has been circled for you. About how many are in one group? _____

b) About how many groups are there in total? _____

c) What is your estimate for the total number of dots? _____

Q2 — Round and Estimate

Round each number to the nearest 10, then estimate the answer.

Problem	Rounded numbers	Estimate
$42 + 36$		
$57 - 21$		
$68 + 19$		

Q3 — Is the Answer Reasonable?

Use estimation to check each answer. Circle Reasonable or Not Reasonable.

Calculation	Given answer	Your estimate	Reasonable?
$39 + 42$	81		Reasonable / Not
$76 - 28$	49		Reasonable / Not
$53 + 19$	172		Reasonable / Not

Watch Out!

An estimate is not the exact answer — it's a quick check. If your exact answer is very different from your estimate, you likely made a mistake and should calculate again.

Q4 — Estimate to Solve a Word Problem

A school hall has 3 rows of chairs with about 28 chairs in each row. About how many chairs are there in total? Show your estimate and explain your thinking.

Apply & Reasoning — Use what you know to explain your thinking.

- 1 Reasoning:** Mo says his answer to $47 + 35$ is 42. Use estimation to explain why this cannot be correct.

- 2 Reasoning:** Why is it useful to estimate before you calculate, not just after?

- 3 Open-ended:** A jar has a collection of marbles. Describe how you would estimate the total without counting every single one. Then invent a total and show it's reasonable using your method.

Quick Check — Exit Ticket

Round 54 and 29 to the nearest 10, then estimate $54 + 29$.

Answer & Teacher Support**Answer Key — Pages 1 & 2**

Try It (p1): $47 \rightarrow 50$; $63 \rightarrow 60$.

Q1 (p2): a) 10 per group. b) About 4–5 groups. c) Estimate about 45–50 (actual dots shown = 47; accept any reasonable estimate in that range with working shown).

Q2 (p2): $42+36 \rightarrow 40+40=80$ (exact 78). $57-21 \rightarrow 60-20=40$ (exact 36). $68+19 \rightarrow 70+20=90$ (exact 87).

Q3 (p2): $39+42 \rightarrow$ est. 80, exact 81 — Reasonable. $76-28 \rightarrow$ est. 50 ($80-30$), exact 48, given 49 is close — Reasonable. $53+19 \rightarrow$ est. 70, exact 72, given 172 — Not Reasonable (likely a place-value error).

Q4 (p2): About $3 \times 30 = 90$ chairs (rounding 28 to 30). Accept any reasonable estimate close to 84 (the exact total) with clear working shown.

Answer Key — Page 3

Q1: Rounding gives $50+40=90$ as an estimate; 42 is nowhere close to 90, so Mo's answer must be wrong (correct answer is 82).

Q2: Estimating first gives you an idea of what size answer to expect, so you can immediately spot a calculation mistake rather than only checking afterwards.

Q3 (acceptable-answer guide): Accept any grouping-based estimation method (e.g. counting a small group and multiplying) applied consistently to a self-chosen total.

Exit ticket: $54 \rightarrow 50$, $29 \rightarrow 30$, estimate $50+30=80$ (exact answer 83, which is close).

Teacher Checklist

- ☐ Estimates a collection using grouping.
- ☐ Rounds numbers to the nearest 10 accurately.
- ☐ Uses an estimate to judge if an exact answer is reasonable.
- ☐ Identifies an unreasonable answer and explains why.
- ☐ Applies estimation to solve a practical word problem.
- ☐ Explains the purpose of estimating before calculating.

Teacher Tip

Watch for students who estimate accurately but then don't use the estimate to check their work — reinforce that estimating is a tool for catching mistakes, not just a separate task to complete.